=== UNIT 1: LINEAR ALGEBRA, MATRICES & EIGENVALUE THEORY ===

>> 1.1 Elementary Matrix Transformations and Matrix Rank

Linear algebra provides the mathematical framework for solving multidimensional physical and biological engineering models.
* Elementary Operations: (1) Interchanging any two rows or columns (Ri <-> Rj). (2) Multiplying any row by a non-zero scalar k (Ri -> k*Ri). (3) Adding a scalar multiple of one row to another (Ri -> Ri + k*Rj). These operations preserve the rank and solution set of associated linear systems.
* Row Echelon Form (REF): A matrix is in REF if: (a) All zero rows are at the bottom. (b) The leading entry (first non-zero number from the left, pivot) of each non-zero row is strictly to the right of the leading entry of the row above it. (c) All entries below each pivot are zero.
* Reduced Row Echelon Form (RREF): An REF matrix where every pivot entry is equal to 1, and each pivot 1 is the ONLY non-zero entry in its entire column.
* Rank of Matrix (rho): Defined as the maximum number of linearly independent row or column vectors. Computationally, rho(A) equals the total count of non-zero rows in its row echelon form. For an m x n matrix, rho(A) <= min(m, n). A matrix of order n with rho(A) = n is non-singular with det(A) != 0.

>> 1.2 System of Linear Algebraic Equations (AX = B)

* Augmented Matrix [A | B]: Formed by adjoining column vector B to coefficient matrix A.
* The Rouche-Capelli Theorem (Fundamental Existence & Uniqueness Theorem):
1. Inconsistent System (No Solution): If rho(A) < rho(A | B). Geometrically, planes/hyperplanes do not intersect at a common manifold.
2. Consistent System with Unique Solution: If rho(A) = rho(A | B) = n (where n is the number of unknown variables). Det(A) != 0.
3. Consistent System with Infinitely Many Solutions: If rho(A) = rho(A | B) = r < n. The system possesses (n - r) independent arbitrary parameters (degrees of freedom).
* Homogeneous Linear Systems (AX = 0): Always consistent since X = 0 (the trivial zero vector) is always a solution. If rho(A) = n, only the trivial solution exists. If rho(A) < n (i.e., det(A) = 0), non-trivial non-zero solutions exist, spanning the null space (kernel) of dimension n - rho(A).

>> 1.3 Eigenvalues, Eigenvectors and Diagonalization

* Characteristic Equation: For a square matrix A of order n, det(A - lambda * I) = 0. Expanding this determinant yields an nth-degree polynomial in lambda. The n roots (real or complex) are the Eigenvalues (latent roots/characteristic values).

* Fundamental Invariants: (1) Sum of eigenvalues = Trace of A (sum of main diagonal elements: sum(a_ii)). (2) Product of eigenvalues = Determinant of A (det(A)). (3) If lambda is an eigenvalue of A, then lambda^k is an eigenvalue of A^k, and 1/lambda is an eigenvalue of A^-1 (if A is invertible).
* Eigenvectors: Non-zero vectors X satisfying (A - lambda * I) * X = 0. Eigenvectors corresponding to distinct eigenvalues are mutually linearly independent.
* Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation. That is, if characteristic equation is p(lambda) = lambda^n + c_{n-1}*lambda^{n-1} + ... + c1*lambda + c0 = 0, then A^n + c_{n-1}*A^{n-1} + ... + c1*A + c0*I = O (zero matrix). Highly efficient for computing higher matrix powers A^m and matrix inverses: A^-1 = (-1/c0) * [A^{n-1} + c_{n-1}*A^{n-2} + ... + c1*I].

[SOLVED PROBLEM 1: Finding Matrix Rank via Row Echelon Form]

Problem: Determine the rank of the 4x4 matrix A by reducing it to Row Echelon Form:
[1 2 -1 3]
A = [2 4 1 -2]
[3 6 3 -7]
[1 2 2 -5]
Solution Step 1: Use pivot a11 = 1 to eliminate entries in column 1 below row 1:
R2 -> R2 - 2*R1: [2-2, 4-4, 1 - (-2), -2 - 6] = [0, 0, 3, -8]
R3 -> R3 - 3*R1: [3-3, 6-6, 3 - (-3), -7 - 9] = [0, 0, 6, -16]
R4 -> R4 - 1*R1: [1-1, 2-2, 2 - (-1), -5 - 3] = [0, 0, 3, -8]
Matrix becomes:
[1 2 -1 3]
A1 = [0 0 3 -8]
[0 0 6 -16]
[0 0 3 -8]
Solution Step 2: Use pivot a23 = 3 to eliminate entries in column 3 below row 2:
R3 -> R3 - 2*R2: [0, 0, 6 - 6, -16 - (-16)] = [0, 0, 0, 0]
R4 -> R4 - 1*R2: [0, 0, 3 - 3, -8 - (-8)] = [0, 0, 0, 0]
Resulting Row Echelon Form:
[1 2 -1 3]
REF= [0 0 3 -8]
[0 0 0 0]
[0 0 0 0]
Conclusion: The number of non-zero rows in REF is exactly 2. Therefore, Rank(A) = rho(A) = 2.

[SOLVED PROBLEM 2: Consistency and Complete Solution of Linear System]

Problem: Investigate the consistency of the system of equations, and if consistent,

solve it completely:
x + y + z = 6
x + 2y + 3z = 10
x + 2y + 4z = 11
Solution Step 1: Construct the augmented matrix [A | B]:
[A | B] = [1 1 1 | 6]
[1 2 3 | 10]
[1 2 4 | 11]
Solution Step 2: Apply elementary row transformations:
R2 -> R2 - R1: [0 1 2 | 4]
R3 -> R3 - R1: [0 1 3 | 5]
Matrix becomes:
[1 1 1 | 6]
[0 1 2 | 4]
[0 1 3 | 5]
Next, eliminate under second pivot:
R3 -> R3 - R2: [0 0 1 | 1]
Final Echelon Form:
[1 1 1 | 6]
[0 1 2 | 4]
[0 0 1 | 1]
Solution Step 3: Evaluate ranks:
rho(A) = 3, rho(A | B) = 3, number of unknowns n = 3.
Since rho(A) = rho(A | B) = n = 3, the system is CONSISTENT with a UNIQUE SOLUTION.
Solution Step 4: Back-substitution:
From Row 3: 1*z = 1 => z = 1.
From Row 2: y + 2*z = 4 => y + 2(1) = 4 => y = 2.
From Row 1: x + y + z = 6 => x + 2 + 1 = 6 => x = 3.
Final Answer: Unique solution is x = 3, y = 2, z = 1.

[SOLVED PROBLEM 3: Complete Eigenvalue and Eigenvector Determination]

Problem: Find all eigenvalues and the corresponding eigenvectors for the matrix:
[8 -6 2]
A = [-6 7 -4]
[2 -4 3]
Solution Step 1: Characteristic equation det(A - lambda * I) = 0:
lambda^3 - S1*lambda^2 + S2*lambda - S3 = 0, where:
S1 = Trace(A) = 8 + 7 + 3 = 18.
S2 = Sum of principal minors:
M11 = det([7 -4; -4 3]) = 21 - 16 = 5.
M22 = det([8 2; 2 3]) = 24 - 4 = 20.
M33 = det([8 -6; -6 7]) = 56 - 36 = 20.
S2 = 5 + 20 + 20 = 45.

S3 = det(A) = 8(21 - 16) - (-6)(-18 - (-8)) + 2(24 - 14) = 8(5) + 6(-10) + 2(10) = 40 - 60 + 20 = 0.
Characteristic Equation: lambda^3 - 18*lambda^2 + 45*lambda = 0.
Factoring: lambda * (lambda^2 - 18*lambda + 45) = 0 => lambda * (lambda - 3) * (lambda - 15) = 0.
Eigenvalues: lambda1 = 0, lambda2 = 3, lambda3 = 15.
Solution Step 2: Eigenvector for lambda = 0:
(A - 0*I)X = 0 => [8 -6 2; -6 7 -4; 2 -4 3][x1; x2; x3] = [0; 0; 0].
From rows: 2*x1 - 4*x2 + 3*x3 = 0 and -6*x1 + 7*x2 - 4*x3 = 0.
Using cross-multiplication: x1 / (16 - 21) = x2 / (-18 - (-8)) = x3 / (14 - 24)
=> x1 / (-5) = x2 / (-10) = x3 / (-10) => x1/1 = x2/2 = x3/2.
Eigenvector X1 = [1, 2, 2]^T.
Solution Step 3: Eigenvector for lambda = 3:
(A - 3*I)X = 0 => [5 -6 2; -6 4 -4; 2 -4 0][x1; x2; x3] = [0; 0; 0].
From row 3: 2*x1 - 4*x2 = 0 => x1 = 2*x2.
Substitute into row 1: 5(2*x2) - 6*x2 + 2*x3 = 0 => 4*x2 + 2*x3 = 0 => x3 = -2*x2.
Let x2 = 1 => x1 = 2, x3 = -2. Eigenvector X2 = [2, 1, -2]^T.
Solution Step 4: Eigenvector for lambda = 15:
(A - 15*I)X = 0 => [-7 -6 2; -6 -8 -4; 2 -4 -12][x1; x2; x3] = [0; 0; 0].
Cross-multiplication of first two rows yields: x1 / (24 - (-16)) = x2 / (-12 - 28) = x3 / (56 - 36)
=> x1 / 40 = x2 / (-40) = x3 / 20 => x1/2 = x2/(-2) = x3/1.
Eigenvector X3 = [2, -2, 1]^T.
Verification: Since A is real symmetric, eigenvectors are mutually orthogonal: X1 . X2 = 2 + 2 - 4 = 0; X1 . X3 = 2 - 4 + 2 = 0; X2 . X3 = 4 - 2 - 2 = 0 (Confirmed).

[SOLVED PROBLEM 4: Verification of Cayley-Hamilton Theorem & Inverse]

Problem: Verify Cayley-Hamilton Theorem for matrix A = [1 4; 2 3] and hence evaluate A^-1 and A^4.
Solution Step 1: Characteristic equation det(A - lambda*I) = 0:
det([1-lambda 4; 2 3-lambda]) = (1-lambda)(3-lambda) - 8 = lambda^2 - 4*lambda - 5 = 0.
Solution Step 2: Verification with Cayley-Hamilton theorem: A^2 - 4*A - 5*I = O:
A^2 = [1 4; 2 3] * [1 4; 2 3] = [1*1+4*2 1*4+4*3; 2*1+3*2 2*4+3*3] = [9 16; 8 17].
4*A = [4 16; 8 12].
5*I = [5 0; 0 5].
A^2 - 4*A - 5*I = [9-4-5 16-16-0; 8-8-0 17-12-5] = [0 0; 0 0] = O. (Verified).
Solution Step 3: Compute A^-1:
Multiply equation by A^-1: A - 4*I - 5*A^-1 = 0 => 5*A^-1 = A - 4*I
5*A^-1 = [1-4 4-0; 2-0 3-4] = [-3 4; 2 -1]
A^-1 = (1/5) * [-3 4; 2 -1] = [-0.6 0.8; 0.4 -0.2].
Solution Step 4: Compute A^4:

From A^2 = 4*A + 5*I, multiply by A: A^3 = 4*A^2 + 5*A = 4(4*A + 5*I) + 5*A = 21*A + 20*I.
Multiply by A: A^4 = 21*A^2 + 20*A = 21(4*A + 5*I) + 20*A = 84*A + 105*I + 20*A = 104*A + 105*I.
104*A = [104 416; 208 312]; 105*I = [105 0; 0 105].
A^4 = [209 416; 208 417].

>> 1.4 University Review & Examination Questions

* Q1: State and prove the Cayley-Hamilton theorem for a general 2x2 matrix.
Answer: Theorem: Every square matrix satisfies its own characteristic equation det(A - lambda*I) = 0. For A = [a b; c d], characteristic polynomial is lambda^2 - (a+d)*lambda + (ad - bc) = 0. We must show A^2 - (a+d)*A + (ad-bc)*I = O. A^2 = [a^2+bc ab+bd; ac+cd bc+d^2]. (a+d)*A = [a^2+ad ab+bd; ac+cd ad+d^2]. Subtracting: A^2 - (a+d)*A = [bc-ad 0; 0 bc-ad] = -(ad-bc)*I. Adding (ad-bc)*I yields [0 0; 0 0] = O. Hence proved.
* Q2: Explain the conditions under which a system of non-homogeneous linear equations AX = B has (i) No solution, (ii) Unique solution, (iii) Infinite solutions.
Answer: By the Rouche-Capelli theorem, let [A | B] be the augmented matrix and n be the number of variables: (i) Inconsistent (No solution) iff rho(A) < rho(A | B). (ii) Consistent with Unique solution iff rho(A) = rho(A | B) = n. (iii) Consistent with Infinite solutions iff rho(A) = rho(A | B) = r < n, yielding n - r arbitrary free parameters.

>> 1.5 Key Takeaways and Revision Points

* Rank is the number of non-zero rows in Row Echelon Form; invariant under elementary row operations.
* Rouche-Capelli theorem governs system consistency: compare rho(A) vs. rho(A | B).
* Sum of eigenvalues equals matrix trace; product equals determinant.
* Cayley-Hamilton theorem allows direct calculation of inverses and polynomial matrix powers without explicit multi-step multiplication.